

Architecture Issues — 50 files · 7,705 lines

CRITICAL

Architecture

God File — Violating SRP

backend/node_modules/@napi-rs/canvas/geometry.js is 874 lines. Files this large do too much and are impossible to unit-test effectively.

- backend/node_modules/@napi-rs/canvas/geometry.js

Fix: Extract into focused modules. Each file should have one reason to change (Single Responsibility).

CRITICAL

Architecture

God File — Violating SRP

backend/node_modules/@napi-rs/canvas/index.d.ts is 734 lines. Files this large do too much and are impossible to unit-test effectively.

- backend/node_modules/@napi-rs/canvas/index.d.ts

Fix: Extract into focused modules. Each file should have one reason to change (Single Responsibility).

MEDIUM

Readability

Function "requireNative" is 186 lines

In backend/node_modules/@napi-rs/canvas/js-binding.js. Functions over 40 lines become hard to reason about and test.

- backend/node_modules/@napi-rs/canvas/js-binding.js

Fix: Break "requireNative" into smaller, well-named helper functions. Each should do one thing.

MEDIUM

Readability

Function "read" is 134 lines

In backend/node_modules/body-parser/lib/read.js. Functions over 40 lines become hard to reason about and test.

- backend/node_modules/body-parser/lib/read.js

Fix: Break "read" into smaller, well-named helper functions. Each should do one thing.

MEDIUM

Readability

Function "json" is 111 lines

In backend/node_modules/body-parser/lib/types/json.js. Functions over 40 lines become hard to reason about and test.

- backend/node_modules/body-parser/lib/types/json.js

Fix: Break "json" into smaller, well-named helper functions. Each should do one thing.

MEDIUM

Readability

Function "expand_" is 103 lines

In backend/node_modules/brace-expansion/dist/commonjs/index.js. Functions over 40 lines become hard to reason about and test.

- backend/node_modules/brace-expansion/dist/commonjs/index.js

Fix: Break "expand_" into smaller, well-named helper functions. Each should do one thing.

MEDIUM

Readability

Function "expand_" is 103 lines

In backend/node_modules/brace-expansion/dist/esm/index.js. Functions over 40 lines become hard to reason about and test.

- backend/node_modules/brace-expansion/dist/esm/index.js

Fix: Break "expand_" into smaller, well-named helper functions. Each should do one thing.

MEDIUM

Logging

7 console statements in backend/node_modules/braces/index.js

Excessive console.log in production code. No structured logging, no log levels, no centralized collection.

- backend/node_modules/braces/index.js

Fix: Replace with a structured logger (Winston, Pino). Use log levels (info, warn, error). Remove debug logs before production.

CRITICAL

Architecture

God File — Violating SRP

backend/node_modules/busboy/lib/types/multipart.js is 654 lines. Files this large do too much and are impossible to unit-test effectively.

- backend/node_modules/busboy/lib/types/multipart.js

Fix: Extract into focused modules. Each file should have one reason to change (Single Responsibility).

MEDIUM

Readability

Function "ssCb" is 161 lines

In backend/node_modules/busboy/lib/types/multipart.js. Functions over 40 lines become hard to reason about and test.

- backend/node_modules/busboy/lib/types/multipart.js

Fix: Break "ssCb" into smaller, well-named helper functions. Each should do one thing.

CRITICAL

Architecture

God File — Violating SRP

backend/node_modules/busboy/lib/utils.js is 597 lines. Files this large do too much and are impossible to unit-test effectively.

- backend/node_modules/busboy/lib/utils.js

Fix: Extract into focused modules. Each file should have one reason to change (Single Responsibility).

MEDIUM

Readability

Function "parseContentTypeParams" is 113 lines

In backend/node_modules/busboy/lib/utils.js. Functions over 40 lines become hard to reason about and test.

- backend/node_modules/busboy/lib/utils.js

Fix: Break "parseContentTypeParams" into smaller, well-named helper functions. Each should do one thing.

MEDIUM

Readability

Function "parseDispositionParams" is 198 lines

In backend/node_modules/busboy/lib/utils.js. Functions over 40 lines become hard to reason about and test.

- backend/node_modules/busboy/lib/utils.js

Fix: Break "parseDispositionParams" into smaller, well-named helper functions. Each should do one thing.

HIGH

Coupling

2 Highly-Coupled Modules

These files import far more dependencies than average (3 avg). High coupling = hard to change, hard to test.

- backend/node_modules/@napi-rs/canvas/js-binding.js (44 imports)
- backend/node_modules/braces/index.js (9 imports)

Fix: Apply Facade pattern to reduce direct dependencies. Use dependency injection. Consider an event bus for cross-module communication.

MEDIUM

Testing

Low Test Coverage

Only 2 test files for 48 source files (~4% ratio).

Fix: Prioritize tests for controllers, services, and data access layers. Use mocks for external dependencies.